

Music Playlist Manager

Problem Description

You are asked to implement a music playlist manager using a doubly linked list. The playlist maintains a *current* pointer that tracks the song currently being played. Initially, the current pointer points to the first song in the playlist.

Each song is assigned a unique id starting from 1 and increasing in the order they are added (including songs in the initial playlist). Deleted ids are never reused. Every newly inserted song via **INSERT** or **ADD** receives a fresh id continuing from the last assigned id.

The playlist supports the following seven operations:

ADD x — Append a new song with name x to the tail of the playlist.

INSERT x — Insert a new song with name x immediately after the current song. The current pointer does not move.

DEL k — Delete the song with id k . If the deleted song is the current song, the current pointer moves to the next song. If there is no next song, the current pointer moves to the previous song. If the playlist becomes empty, the current pointer is set to NULL.

NEXT — Move the current pointer to the next song. Ignored if the current song is already the last song.

PREV — Move the current pointer to the previous song. Ignored if the current song is already the first song.

PRINT — Print the names of all songs from head to tail, separated by spaces. Print **EMPTY** if the playlist is empty.

NOW — Print the name of the current song.

Input Format

The first line contains two integers N and Q .

Each of the next N lines contains the name of an initial song, in order from head to tail.

Each of the next Q lines contains one operation in the format described above.

Notes:

- The initial songs are assigned ids 1 to N in order. New song ids start from $N + 1$ and increase in the order they are inserted.
- **DEL** only moves the current pointer if the deleted song is the current song; otherwise the current pointer remains unchanged. All ids referenced by **DEL** are guaranteed to be valid at the time of the operation.
- If the playlist is empty when **INSERT** is called, the new song becomes the only song in the playlist and the current pointer is set to it. The same applies to **ADD**.
- Song names consist of printable **non-space** ASCII characters with length between 1 and 20.

Output Format

For each **PRINT** or **NOW** operation, output one line as described above.

Constraints

- $1 \leq N, Q \leq 2 \times 10^5$
- Time Limit: 1 second; Memory Limit: 256 MB.

Example Test Case

Sample Input 1

```
3 8
Dynamite
Butter
Fireflies
NOW
NEXT
NOW
PREV
NOW
PRINT
INSERT Radioactive
PRINT
```

Sample Output 1

```
Dynamite
Butter
Dynamite
Dynamite Butter Fireflies
Dynamite Radioactive Butter Fireflies
```

Sample Input 2

```
3 9
Believer
Levitating
Titanium
NEXT
NOW
DEL 2
NOW
PRINT
DEL 3
PRINT
DEL 1
PRINT
```

Sample Output 2

```
Levitating
Titanium
Believer Titanium
Believer
EMPTY
```

Sample Input 3

```
2 8
Psycho
Ditto
NEXT
INSERT Lilac
PRINT
DEL 2
NOW
PREV
NOW
PRINT
```

Sample Output 3

```
Psycho Ditto Lilac
Lilac
Psycho
Psycho Lilac
```